Soring Kubernetes

Some tools in this space.

* Kubernetes
  ↳ K8s is a container orchestration engine and runtime. While it has become defacto standard for production container runtime environments.

* kind
  ↳ Kubernetes in Docker (kind) was developed by the kubernetes sigs (special interest group) to test Kubernetes itself but alloces you to run Kubernetes instance on your local development machine.

* Helm
  * templatize various kubernetes manifest needed to deploy application of various types -
  * Raw Kubernetes manifests needed to deploy applications of varias type.

* Postgress SQL:
  open saurce object relational type relational database management

system (RDBMS) that powers
world's relational db work

Kubernetes :- * open source container orchestration platform launched by Google and donated to Cloud Native Computing Foundations.

* Planet scale deployments, allowing both small & large companies.

* Automatic scaling & elasticity

* run on any server, premises in cloud or both.

Containers : Bundle the code and config required to run application in one unit making them portable across different OS.

Advantages :-

* less CPU and memory
* virtual machines
* container images are executable files that can be run as containers stored in container registration like Docker Hub.
* Kubernetes pulls container images

from registries, manages the number of replicas and containers runtime interface to build and start containers.

* Cloud Native Technologies empower organizations to build and run scalable applications in dynamic environments like public, private and hybird clouds.

* CNCF supports all this.

* Kubernetes is a cloud-native technology supported by CNCF designed to automate the deployment and scaling of applications and has become the de facto container or....

→ Minikube runs locally on your computer where you can run k8 cluster.

* ensure docker is running.

* minikube start

* kubectl

* kube ctl get nodes

* kube ctl get namespaces

* kubectl get pods -A

* kubectl get services -A

YAML

YAML Ain't Markup Language

--- ← start of a document

```
# my record
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 cours:
      - Learning K8S
      - Spring

 jobs:
      Digital Ocean:  1.2 3
```

- YAML is a data seridization language often used for configuration files in Kubernetes.

- YAML is human-readable, support key-value pairs, sequences (lists) and nested structures.

* K8s proces YAML files. These files are used to define the desired state of your kubernets cluste.

* Name spaces help isolate and organize workloads in Kubernetes _

* Default Name spaces ⟶ 4 default Namespacy.

* Creating name spacs ⟶ use YAML manifests to create namespaces and apply them with kubectl apply -f namespace .yml.

* Managing Multiple Resources: .
Define multiple resources in different yaml files by separating them with three dashes (- - -)

* Deleting namespaces

kubectl delete -f namespac.yml

kubernetes ensures high availability by running multiple replicas of your applicaton.

If one pod stops working, others will continue to accept trafic.

kubernetes Deployment:

A Deployment is a kuberyntes object that manages a group of pods. You can speaty the numbir of replicas and other configurations in .YAML file.

Creating a Deployment

→ Use kube ctl apply -f deployment.yml command to create a deployment.

Trouble shooting pods: Event logs are quite crucial for identifying issues that occur when a pod is crashed.

Common problems include unavilable container images lack of a space on worker nodes.

The kubectl describe pod command is used to view detailed information about pod, including event logs.

Event Logs: Event logs show message that help identify problems or confirm that a pod is running healthily.

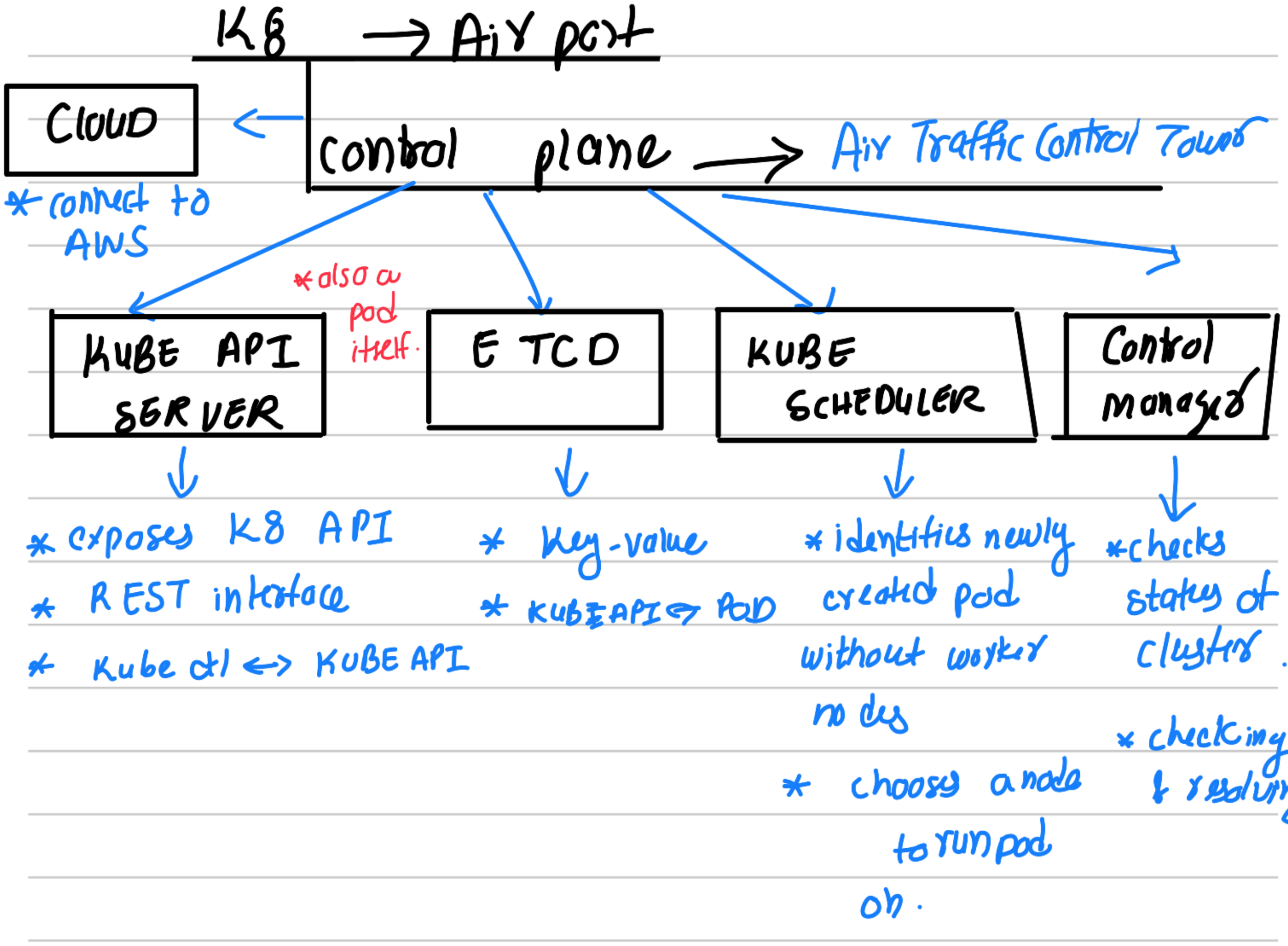
* Busy Box Tool: versatile tool containing many Unix utilities, useful for debugging and trouble shooting in a Linux environment

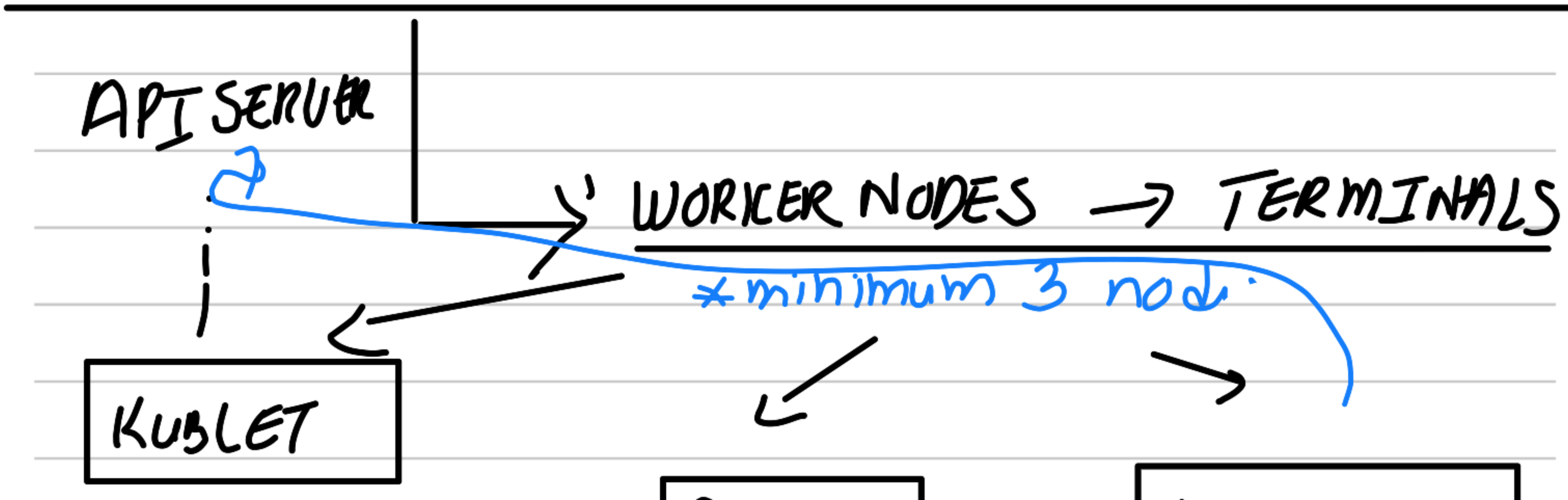
* Creating a busy box pod: You can create a busy box pod using a yaml file and the kubectl apply -f busybox.yaml command.

* Verifying Application: Use BusyBox to make an HTTP GET request to your application by finding the pod's IP Address and using wget command.

* Using Kubectl exec to run commands inside container.

→ Kubernetes cluster : An instance of K8 is called a cluster which includes a control plane + atleast one worker node.





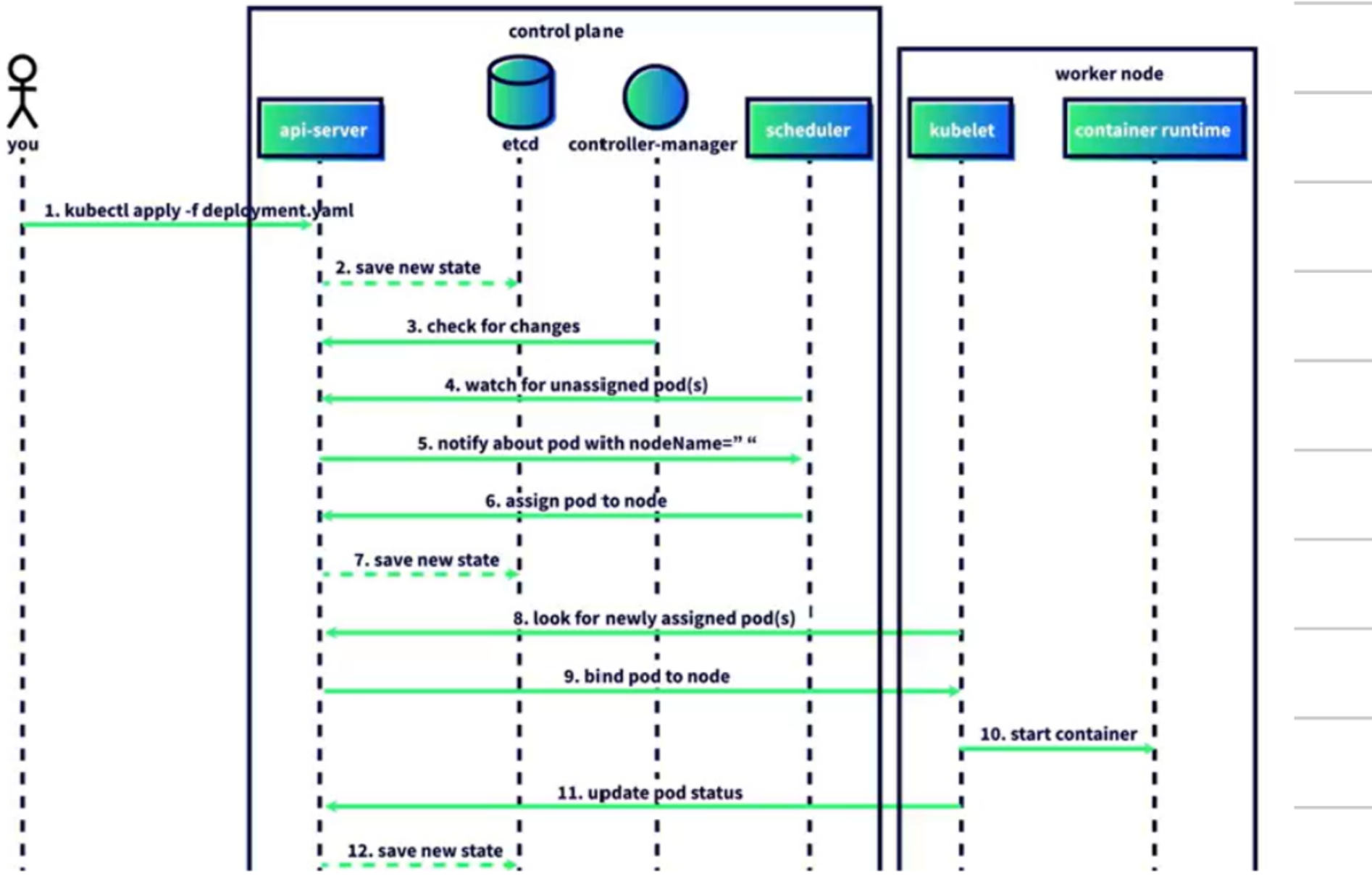
* An agent that ensures that containers in a pod run healthy.

Container Runtime

starts containers.

Kube Proxy

communication between pods and services.



Other ways to manage pods :-

→ Replicas
→ No down time upgrade.

DAEMON SETS

→ One pod per node
→ Background processes.